Base document d
“Skai TV is a Greek television station headquartered in Piraeus.”
Entity
Skai TV
Type
television station
Location
Piraeus
Extraction configuration θ
Model: DeepSeek-V4-Flash
Schema: (country, located_in, headquarters_location)
Prompt: default IE prompt
Evidence: original order
Decoding: T=0.0, seed=7
Configuration fingerprint
Base graph view G
(Skai TV, country, Greece)
(Skai TV, headquarters_location, Piraeus)
country
Greece
Skai TV
headquarters_location
Piraeus
Reference materialization
$\mathcal{K}_1$ Schema form
Controlled Perturbation
Rename relation labels
Reorder the schema
Observed graph drift
(Skai TV, country, Greece)
(Skai TV, located_in, Piraeus)
$\mathcal{K}_2$ Prompt form
PROMPT
Controlled Perturbation
Paraphrase the instructions
“Document first vs. schema first”
Observed graph drift
drop headquarters_location edge
(Skai TV, country, Greece)
$\mathcal{K}_3$ Evidence form
Controlled Perturbation
Reorder evidence sentences
Substitute entity aliases
Observed graph drift
Skai TV
Piraeus
Greece
(Skai TV, country, Greece)
(Skai TV, headquarters_location, Greece)
$\mathcal{K}_4$ Query-answer stability
Controlled Perturbation
Q3: banks that acquired Swiss-headquartered companies
(Apex Bank, acquired, MedSure)
(MedSure, headquarters_location, Basel)
(Basel, country, Switzerland)
Observed answer drift
Base: {Apex Bank}
After: ∅
$\mathcal{K}_5$ Model replacement
Controlled Perturbation
DeepSeek
Qwen
Kimi
Possible result
Model A: headquarters_location(Piraeus)
Model B: located_in(Piraeus)
Model C: edge dropped
$\mathcal{K}_6$ Decoding resample
Controlled Perturbation
Run 1/ Run 2/ Run 3
same θ, T=0.3, seed=7/13/23/37/53
Possible result
Run 1: headquarters_location(Piraeus)
Run 2: located_in(Piraeus)
Run 3: edge dropped
Run 1
Run 2
Run 3